

Running the Course VM with QEMU

Getting the image, and the two mistakes everyone makes.

Getting the image

QEMU needs the disk image to boot. There are two ways to get it.

Option 1. Follow Tomer's guide.

Option 2. Use the copy already sitting in your /scratch directory. Every account has its own: ece882-001 has /scratch/ece882-001 on naranja10, and it runs up to ece882-039 with /scratch/ece882-039 on naranja7.

Each account maps to a specific server, listed in the CSV. Log in to that server, not another one, or the /scratch copy will not be there.

Running it

Swap in your own account and the server from the CSV:

```
% ssh ece882-001@naranja10

ece882-001@naranja10:~$ cd /scratch/ece882-001

ece882-001@naranja10:/scratch/ece882-001$ qemu-system-x86_64 -m 4096m \
-nographic -cpu host -accel kvm \
-nic user,model=virtio-net-pci \
-drive file=jammy-server-cloudimg-amd64-disk-kvm.img,format=qcow2
```

The cd matters. The image lives in /scratch, so run QEMU from there. If you run it somewhere else without a full path to the image, QEMU has nothing to boot and it fails.

Mistake 1: the image is not where QEMU is looking

A command like the one below fails when the image is not in the current directory or you have not given a valid path to it:

```
qemu-system-x86_64 -m 4096m -nographic -cpu host -accel kvm \
-nic user,model=virtio-net-pci \
-drive file=jammy-server-cloudimg-amd64-disk-kvm.img,format=qcow2
```

Either cd into the directory that holds the image, or point -drive at the full path.

Mistake 2: building on the host instead of in the VM

gcc errors and similar build failures usually mean you are compiling on the server you logged into. All the work happens inside the VM. Start QEMU, wait for the guest to boot, log in to the guest, and build there. The errors go away once you are actually inside.

When something breaks, the first question is which machine you are on: the host or the guest. The prompt and the error usually tell you.